Task 6 - System-call cost comparison

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main.c

```c
    for (int i = 0; i < 10000; i++)
        fputc('A' + (i % 26), f);
    fclose(f);

    size_t sizes[] = {1, 512, 4096};
    for (int i = 0; i < 3; i++) {
        char dst[32];
        snprintf(dst, sizeof dst, "copy_%zu.txt", sizes[i]);
        long calls = copy_file(src, dst, sizes[i]);
        printf("buffer %4zu bytes: %ld read()/write() calls\n",
               sizes[i], calls);
    }
    return 0;
}
```

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Output Finished

```
$ main.c arguments (optional)

buffer    1 bytes: 20001 read()/write() calls
buffer  512 bytes: 41 read()/write() calls
buffer 4096 bytes: 7 read()/write() calls


** Process exited - Return Code: 0 **
```

Task 7 - Process control with fork, exec, and wait

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main.c

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main(void) {
    pid_t first = fork();

    if (first < 0) {
        perror("fork");
        return 1;
    }

    if (first == 0) {
        printf("Child 1: PID=%d, PPID=%d\n", getpid(), getppid());
```

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Output Finished

```
$ main.c arguments (optional)

Child 1: PID=21913, PPID=21907
Parent: child 21913 exited with status 0
Child 2: PID=21914, PPID=21907
Parent: child 21914 exited with status 3
```

Task 10 - Process versus thread timing

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main.c

```c
#define _POSIX_C_SOURCE 200809L
#include <pthread.h>
#include <stdio.h>
#include <stdlib.h>
#include <sys/mman.h>
#include <sys/wait.h>
#include <time.h>
#include <unistd.h>

#define N 10000000L
```

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Output • Finished

```
$ main.c arguments (optional)

Environment: Linux POSIX; workload=10000000 increments
Process run 1: 0.196628 seconds, counter=13914021
Process run 2: 0.202546 seconds, counter=13062856
Process run 3: 0.383836 seconds, counter=14461296
Thread run 1:  0.219501 seconds, counter=17123478
Thread run 2:  0.294725 seconds, counter=17085635
Thread run 3:  0.198540 seconds, counter=16249131


** Process exited - Return Code: 0 **
```